

# Hasan Haider

Software Engineer

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## Technical Skills

**Languages:** Python, TypeScript, JavaScript, SQL, Swift, C/C++, C#, HTML/CSS

**Frameworks:** React, React Native, Node.js, Django, SwiftUI, Tailwind CSS, Three.js

**Data & Infrastructure:** MySQL, Supabase, ChromaDB, Cloudflare Workers, WebSockets, SSE

**Tools:** Git, GitHub Actions, Linux, Playwright, Selenium, PyInstaller, Ollama, OpenAI API

## Experience

**Community Dreams Foundation** | *Full Stack Developer*

Jul 2026 - Present

- Develop a workforce training platform with role-specific courses, certifications, quizzes, and progress tracking using React, TypeScript, Tailwind CSS, Supabase, and Bun.
- Implement role-based access, user registration, and administrator approval; build AI-assisted quizzes and workforce analytics for the certification portal.

**GAE Systems** | *Software Engineering Intern*

May 2025 - Aug 2025

- Rebuilt a Python thermal-conductivity instrument application, reducing load time by **38%** and active threads by **25%** through modular design and revised threading logic.
- Built a thread-safe live data plotter, automated error handling, and a PyInstaller pipeline for Windows distribution.

**GeoAnalysis Engineering** | *Frontend React Developer*

Jul 2023 - Jun 2024

- Replaced a legacy website with a responsive React and Tailwind CSS implementation based on client requirements, improving loading performance by **55%**.
- Refactored frontend components and adapted layouts for desktop, tablet, and mobile screens.

## Selected Projects

**Earth Orbit Live** TypeScript, React, Three.js, Cloudflare Workers

- Built a 3D satellite explorer with search, orbit paths, ground tracks, and city pass predictions; moved orbital propagation and pass calculations to Web Workers.
- Implemented a CelesTrak data proxy with validation, caching, stale-data fallback, and retry backoff; added regression tests for orbital calculations, playback, and proxy behavior.

**Motion Lab** React, Node.js, Three.js, WebSockets, SSE

- Turned phone motion sensors into a 3D game controller with QR pairing, calibration, hit detection, and vibration feedback.
- Built a Node.js relay with WebSocket streaming, HTTP fallback, SSE delivery, session expiry, and payload limits.

**macmpv** Swift, SwiftUI, libmpv, FFmpeg

- Built a native macOS media player with hardware-accelerated playback, local and network media, subtitles, and a reorderable playback queue.
- Integrated fprobe metadata, saved playback positions, seek previews, and configurable subtitle appearance.

**Adaptive Multimedia Edubot** Python, Django, MySQL, Ollama, ChromaDB

- Led a six-person team extending an adaptive tutor with Wikimedia image retrieval, semantic image caching, and local LLM support through Ollama.
- Contributed shared Ollama/OpenAI integration and guided learning workflows; coordinated development through Taiga and weekly scrum meetings.

## Education

**Arizona State University**

M.S. Software Engineering, Cybersecurity Specialization

Aug 2024 - May 2026

GPA: 3.5/4.0

**KIET Group of Institutions**

B.Tech Information Technology

Aug 2019 - Jul 2023

GPA: 7.32/10

## Selected Publication

Co-author, "Neural Network Approaches for Computation of Soil Thermal Conductivity." *Mathematics*, 10(21), 3957 (2022). DOI: 10.3390/math10213957.